



# Design-to-cost for TRB-TRB wind turbine main bearing systems

J. T. Terwey<sup>1</sup> · J. Mendelin<sup>1</sup>

Received: 29 September 2024 / Accepted: 29 January 2025

© Der/die Autor(en), exklusiv lizenziert an Springer-Verlag GmbH Deutschland, ein Teil von Springer Nature 2025

## Abstract

Wind energy is a crucial component of the global transition to renewable energy, and enhancing the performance, reliability, and cost-efficiency of wind turbines is key to their widespread adoption. One of the most important aspects of turbine reliability is the design of the main bearing system, which supports the drivetrain and experiences significant mechanical loads. This paper presents an in-depth analysis of the TRB-TRB (tapered roller bearings in O-configuration) main bearing system, focusing on how variations in roller diameter, load angle, and bearing geometry affect load distribution, contact pressure, rating life, and overall cost efficiency.

Using Finite Element analysis from thyssenkrupp Rothe Erde, the study systematically examines a generic 6MW onshore turbine drivetrain with roller diameters ranging from 60 to 100mm and load angles between 5 and 40°. The results demonstrate that optimized configurations significantly reduce manufacturing costs while maintaining technical integrity. These findings highlight the effectiveness of early-stage design optimization for achieving better cost efficiency, in line with the “Design-to-Cost” approach.

By combining technical simulation with economic analysis, the study provides a comprehensive framework for improving wind turbine main bearing systems. The insights from this research are particularly valuable for designing more efficient and cost-effective turbines, especially as turbines grow in size and capacity. Although the primary focus is on onshore wind turbines, the methods and findings could also be applied to offshore environments, where further research could explore the impact of harsher operational conditions on bearing performance.

## Design-to-Cost für TRB-TRB-Hauptlagersysteme in Windenergieanlagen

### Zusammenfassung

Windenergie ist ein zentraler Bestandteil des globalen Übergangs von fossiler zu erneuerbaren Energien. Die Steigerung der Leistung, Zuverlässigkeit und Kosteneffizienz von Windenergieanlagen (WEA) ist daher ein entscheidender Faktor für ihre breite Nutzbarkeit und Etablierung. Ein wesentlicher Faktor für die Steigerung der Zuverlässigkeit von WEA ist das Design der Hauptlagerung im Triebstrang, welche erheblichen mechanischen Belastungen ausgesetzt ist. Diese Studie stellt eine detaillierte Analyse eines TRB-TRB-Hauptlagersystems (angestellte Kegelrollenlager in O-Anordnung) vor und untersucht, wie Variationen des Rollendurchmessers, des Druckwinkels und der Lagergeometrie die Lastverteilung, Laufbahnpressung und Lebensdauer auf der einen, und die Wirtschaftlichkeit auf der anderen Seite beeinflussen. Unter Verwendung eines Finite-Elemente-Ansatzes von thyssenkrupp Rothe Erde wird dazu ein generischer 6-MW-Antriebsstrang einer Onshore-WEA systematisch untersucht. Dabei werden Rollendurchmesser von 60 bis 100 mm und Druckwinkel zwischen 5 und 40° analysiert. Die Ergebnisse zeigen, dass technisch-wirtschaftlich optimierte Hauptlager-Konfigurationen die Herstellungskosten erheblich senken können, ohne die technische Integrität zu beeinträchtigen. Die Erkenntnisse unterstreichen die Bedeutung einer frühzeitigen Designoptimierung zur Verbesserung der Kosteneffizienz im Sinne eines „Design-to-Cost“-Ansatzes. Durch die Kombination technischer und wirtschaftlicher Analysen bietet die Studie einen Rahmen zur Optimierung von Hauptlagersystemen in WEA. Die gewonnenen Erkenntnisse sind anwendbar für die Entwicklung effizienterer und kostengünstigerer WEA, insbesondere angesichts deren zunehmender Größe und Leistung. Obwohl der Schwerpunkt der Studie auf Onshore-WEA liegt, können die Methoden und Ergebnisse auch auf Offshore-Anwendungen übertragen werden.

✉ J. T. Terwey  
torben.terwey@thyssenkrupp-rotheerde.com

<sup>1</sup> thyssenkrupp rothe erde Germany GmbH, 59555 Lippstadt, Germany

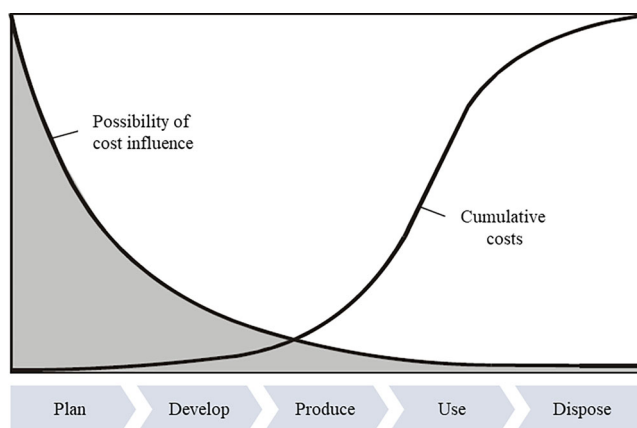


Fig. 1 Product Life Cycle and corresponding costs [2]

## 1 Introduction

As global efforts to reduce carbon emissions intensify, renewable energy sources, particularly wind energy, have gained significant traction. Wind turbines are central to this movement. However, optimizing their performance and reducing invest costs remain critical challenges. The main bearing system of a wind turbine is one such component that requires careful consideration due to its role in ensuring the mechanical integrity and efficiency of the turbine. Main bearings in wind turbines are subjected to substantial loads, making their design and material composition critical for the overall reliability and cost of the turbine. Wind turbines have evolved significantly over the past decades, with modern designs focusing on maximizing energy capture while minimizing operational (OpEx) and capital expenditures (CapEx).

The basic components of a wind turbine include the rotor blades, hub, nacelle, and tower. The rotor blades capture wind energy, which is transferred to the drivetrain housed in the nacelle. The efficiency and reliability of this drivetrain, particularly of the main bearing system, are paramount to the turbine's performance. This paper explores increasingly popular adjusted tapered roller bearing configurations, comparing its technical and economic attributes to identify the optimal system for wind turbine applications.

Reducing both CapEx and OpEx of a turbine makes it even more important to consider cost implications early in the design process and associate it with long-term technical integrity and reliability. Figure 1 illustrates this “Design-to-Cost” approach, showing the impact of early cost analysis in product development. It is divided into five phases: Plan, Develop, Produce, Use, and Dispose. The graph presents two curves: the left one shows that the potential for influencing costs is highest in the early phases and decreases as the project progresses. The right one indicates that making changes becomes increasingly complex and costly as

the project nears production and use. Overall, the graph highlights the need for early cost consideration to manage expenses efficiently throughout the development process.

## 2 Literature review

Major studies underscore the critical importance of early-stage cost analysis, comprehensive drivetrain evaluation, and the integration of economic and ecological factors in wind turbine design. They emphasize the trade-offs between different drivetrain configurations, the impact of technological advancements in reducing costs, and the necessity for holistic approaches in decision-making to optimize performance, cost-effectiveness, and sustainability in wind energy technology.

Polinder et al. [8] compare five different generator systems for wind turbines, including doubly-fed induction generators (DFIG) with three-stage gearboxes and direct-drive permanent-magnet generators (DDPMG). The study finds the DFIG system to be the most cost-effective due to its use of standard components, despite lower energy yield because of significant gearbox losses. Direct-drive systems are more efficient but heavier and more expensive. Among the configurations, a specific DFIG emerges as the best option for balancing energy yield and cost, largely due to lower converter rating requirements, which reduces both cost and losses. This work highlights the complex trade-offs between cost and efficiency in generator selection.

McMillan and Ault [7] evaluate direct drive versus gearbox-driven wind turbines, assessing their technical and economic merits. The study investigates whether direct-drive turbines, which theoretically experience fewer mechanical failures, can provide better reliability and economic performance compared to traditional gearbox-driven turbines. Using a Markov chain model and Monte Carlo simulations, the analysis focuses on key components such as the gearbox, generator, rotor blades, and electronic controls. Results indicate that direct-drive turbines show only marginally better availability due to fewer mechanical failures. However, gearbox-driven turbines currently offer higher revenue, attributed to lower repair and replacement costs. The study concludes that direct-drive systems' economic disadvantage lies in their higher generator failure rates and costs. Nevertheless, it suggests that improvements in the reparability of direct-drive generators and reductions in their component costs could enhance their competitiveness in the future.

Carroll et al. [1] examine the performance, availability, and operation and maintenance (O&M) costs of four prominent offshore wind turbine drivetrain configurations. The study utilizes failure rate data from onshore and offshore turbine populations to estimate O&M costs and availability for a baseline wind farm. Results indicate that turbines with

permanent magnet generators (PMG) and full converters exhibit higher availability and lower O&M costs compared to DFIG, regardless of their proximity to shore. Within PMG configurations, direct-drive turbines show the highest availability and lowest O&M costs, followed by turbines with two-stage and three-stage gearboxes. The elimination of the gearbox in direct-drive configurations significantly reduces failures and downtime, highlighting their cost-effectiveness.

Reder et al. [9] address the importance of understanding wind turbine component failures for developing cost-effective O&M strategies. The study identifies three main challenges in wind turbine failure analysis: non-uniform data treatment, a lack of available failure analyses, and underutilization of alternative data sources such as SCADA alarms. It introduces a modernized taxonomy for classifying wind turbine failures, verified using a database of over 4300 turbines, revealing that direct-drive turbines experience significantly fewer failures and downtime than geared turbines. The study also explores the potential of using SCADA systems to supplement failure logs, suggesting that these systems can predict and prevent component failures effectively.

Guo et al. [4] investigate the impact of drivetrain configurations on wind turbine mass and capital cost, comparing three-point and four-point setups for turbines ranging from 1.5 MW to 5.0 MW. Using the Drive Systems Engineering (DriveSE) tool, the study analyzes, and sizes major drivetrain components based on rotor loads. Key findings reveal that a three-point configuration results in a 16.7% reduction in total nacelle mass and an average 3.5% decrease in turbine capital cost compared to a four-point configuration. However, the study notes that the three-point configuration's lighter design involves a trade-off with potential reliability benefits offered by four-point configurations, which can enhance gearbox reliability by isolating non-torque loads.

Harzendorf et al. [5] present a cost estimation method for wind turbine drivetrains during the early design phase, addressing the economic pressures on wind turbine technology. The method breaks down the drivetrain into key components, such as the gearbox, generator, main shaft, main frame, and bearing arrangement, and uses a bottom-up approach to analyze the manufacturing processes for each component. By focusing on geometric component specifications, the method simplifies input requirements. The study compares its cost estimates to industry benchmarks using a 5 MW wind turbine drivetrain as a reference, finding accurate results that are slightly lower than industry standards due to the exclusion of commercial markups. A case study comparing two gearbox designs for 5 MW and 6.5 MW turbines illustrates the trade-offs between weight optimization and manufacturing costs, highlighting the importance of considering these factors during early design stages. The

paper emphasizes that early manufacturing cost analysis is crucial for balancing innovation with practical manufacturing constraints, ultimately improving the performance and cost-effectiveness of wind turbine drivetrains.

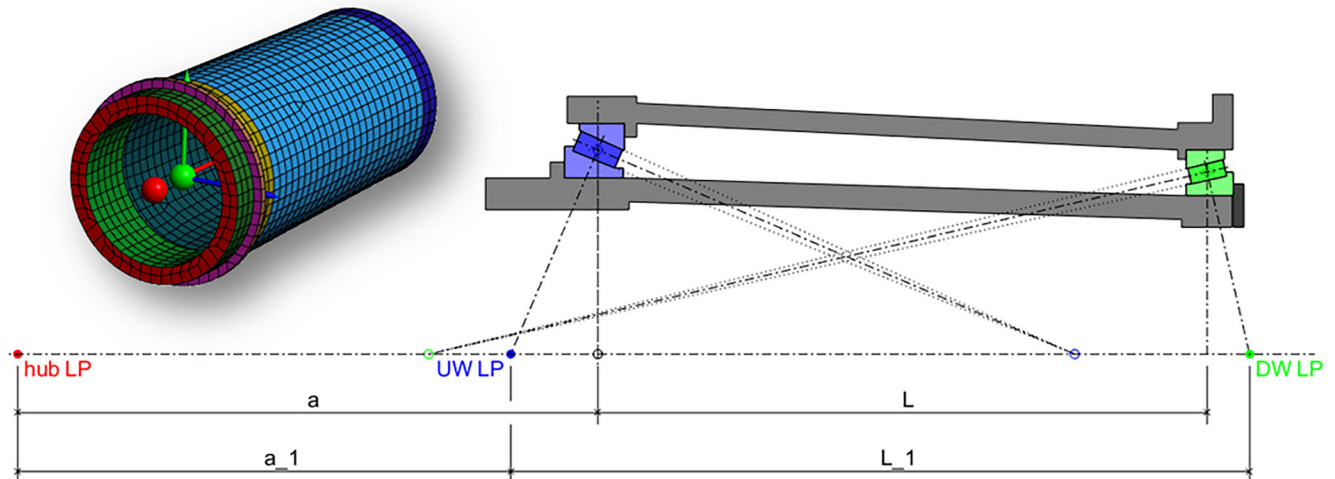
Frumușanu et al. [3] tackle manufacturing cost estimation for roller bearings, proposing a method based on causal identification and comparative assessment to enhance efficiency. The approach focuses on determining key variables like the bearing's exterior and interior diameters, dynamic capacity, and weight, reducing the number of variables involved in the estimation process. The method's comparative assessment involves nonlinear regression to rank potential manufacturing options based on past case data. A case study validates the approach, demonstrating successful reduction of cause-variables from ten to six without compromising estimate quality. This method simplifies the estimation process and delivers reliable cost estimates more efficiently, optimizing roller bearing manufacturing.

Stehly and Beiter [10] provide an analysis of the levelized cost of energy (LCOE) for both on- and offshore wind projects in the United States, using 2018 data to evaluate cost trends. For land-based wind energy, the study estimates an LCOE of \$ 42/MWh for 2.4 MW turbines, attributing the decline in costs to technological advancements, larger turbine sizes, and operational efficiency improvements. Offshore wind energy, analyzed for fixed-bottom and floating turbines using 5.5 MW models, shows LCOEs of \$ 89/MWh and \$132/MWh, respectively, due to higher capital and operational expenditures. The report highlights the significance of CaPex in influencing LCOE and suggests future cost reductions will result from technological innovations and economies of scale.

Harzendorf et al. [6] present a holistic lifecycle evaluation approach for wind turbine drivetrains, comparing geared and direct-drive systems. Using a physics- and empirically-based model, the study evaluates these systems across various lifecycle stages, including investment, operational, and end-of-life phases. The research introduces the "drivetrain specific energy supply effort" (DSE) metric, incorporating external factors like CO<sub>2</sub> emissions to offer a comprehensive evaluation beyond traditional cost metrics such as LCOE. The findings suggest that both geared and direct-drive systems have similar lifecycle costs, with geared systems performing better in strong wind conditions and direct-drive systems in weaker wind environments.

### 3 Methodology

The methodology for this study integrates computational modeling and cost analysis to evaluate the technical performance and cost-efficiency of the TRB-TRB main bearing system used in wind turbines. The approach is divided into



**Fig. 2** Sketch of TRB main bearings system with significant system parameters. Top left: FE model of the shaft, modeled in MESYS, with coordinate system and load introduction point (red ball/red surface)

two main components: technical evaluation and cost assessment. These components aim to provide a comprehensive understanding of how variations in design parameters affect both the mechanical and economic performance of the bearing system.

### 3.1 Technical evaluation

The technical evaluation was conducted using a Finite Element simulation tool chain developed by thyssenkrupp Rothe Erde and integrates the commercial software MESYS. Boundary conditions as fixed housing, rotating shaft, and load application via the hub were applied to replicate realistic operating conditions. Figure 2 shows the main bearing system and FE model used in this study. The simulation tool chain was applied to model the TRB-TRB main bearing configuration within a two-point integrated drivetrain system, as shown in Fig. 3a.

The simulation replicates typical operating conditions for 6MW onshore class wind turbines, including axial and radial forces, moments, rotational speeds, and environmental factors such as temperature and wind loads. The analysis utilized a Damage Equivalent Load case (DEL). The main bearings used in the simulation have pitch circle diameters in a range from approximately 1500 (downwind) to 1750 mm (upwind), aligning with the size requirements for large-scale onshore wind turbine applications.

To thoroughly assess the system's performance, various roller diameters (ranging from 60 to 100 mm) and load angles (from 5 to 40°) were evaluated. These variations were selected to explore their influence on key parameters, including load distribution, maximum raceway pressure, and rating life of the bearing. The following technical aspects were analyzed:

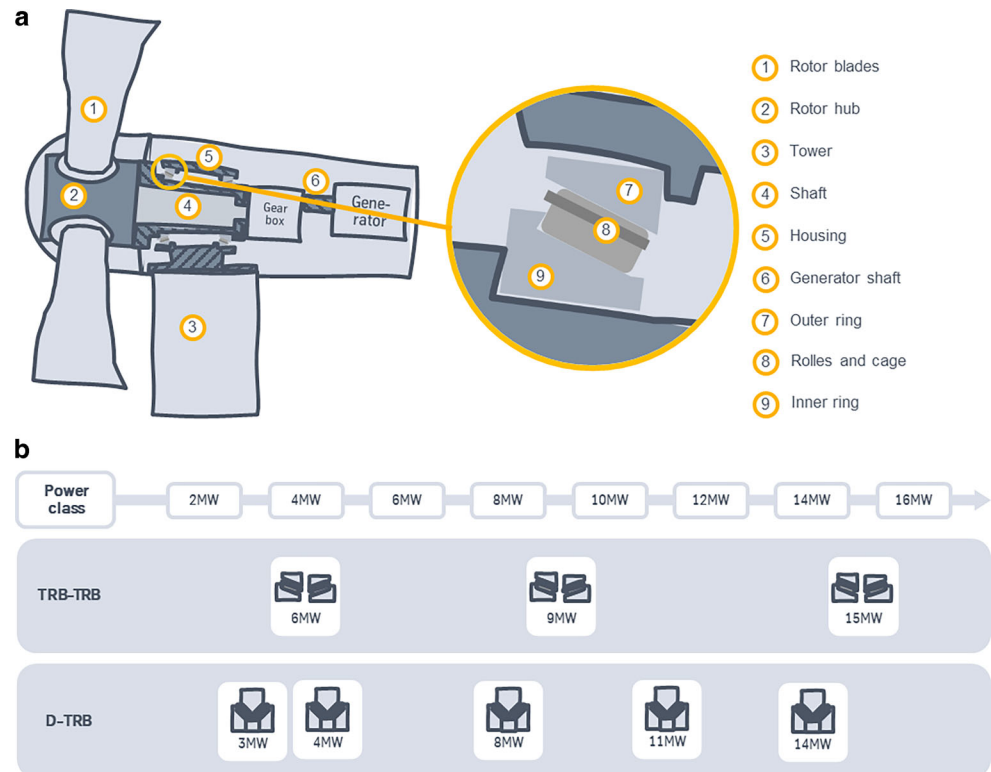
- **Load Distribution:** The simulation calculated the load distribution within the bearings. This distribution directly affects the rating life and technical performance of the bearing system, with optimized designs expected to minimize uneven load distribution.
- **Contact Pressure:** By varying the roller diameter and load angle, the simulation aimed to identify configurations that reduce the maximum contact pressure between the rollers and raceways.
- **Rating Life:** The bearing's reference rating life  $L_{10r}$ , calculated based on ISO standard, was assessed for each configuration. The expected rating life was compared across various geometrical configurations, providing insights into the relationship between design parameters and reliability.

### 3.2 Cost assessment

In parallel with the technical evaluation, a detailed cost analysis was performed using a mass-specific cost model. This model correlates the mass of the bearing components—such as rollers, inner and outer rings, and cages—with manufacturing and material costs. The objective of this analysis was to determine how changes in design parameters (i.e., roller diameter and load angle) impact the total cost of producing the bearing system. The following steps were taken to conduct the cost analysis:

- **Component Mass Calculation:** The mass of each key component (rollers, inner rings, outer rings, and cages) was calculated for different design configurations. This calculation considered variations in roller diameter, load

**Fig. 3** **a** Schematic representation of a fully integrated two-point TRB-TRB wind turbine drive train. **b** TRB-TRB and DTRB systems used to derive bearing components' cost factors



angle, and overall geometry, which directly influence the weight of the bearing assembly.

- **Cost Modeling:** Manufacturing costs for each component were estimated based on industry-standard practices. Constant cost factors (denoted as “c”) were derived to establish the relationship between component mass and manufacturing cost for rollers, inner and outer rings, and cages. These cost factors were then applied to calculate the total production cost for each design configuration.
- **Optimization Process:** The results of the technical analysis were combined with the cost modeling to identify cost-optimized bearing designs. This process involved systematically adjusting roller diameter and load angle to strike a balance between minimizing maximum contact pressure, extending rating life, and reducing manufacturing costs.

Consequently, for both bearings of the main bearing system (upwind and downwind), the overall costs were estimated by multiplying the constant cost factor for the rollers, rings, and cage by their respective weight and adding them together, as outlined in Eq. 1. The total costs are the sum of the individual costs of both bearings, as expressed in Eq. 2. Other effects from shaft, housing and other surrounding components were neglected. The constant cost factors were derived from a detailed analysis of TRB-TRB and Double-row Tapered Roller Bearing (DTRB) systems across wind turbines ranging from 3 to 15MW, as depicted in Fig. 3b.

This evaluation ensured that the cost factors reflect the actual material and manufacturing complexity for bearings used in both lower- and higher-capacity turbines, allowing for a more accurate and scalable cost estimation. By incorporating these factors, the study provides a robust cost model that can be applied to various turbine sizes, ensuring that the optimization of bearing designs is aligned with both technical performance and economic feasibility.

$$C_{UW/DW} = \underbrace{m_{ro} \cdot c_{ro}}_{\text{rollers}} + \underbrace{m_{IR} \cdot c_{IR}}_{\text{inner ring}} + \underbrace{m_{OR} \cdot c_{OR}}_{\text{outer ring}} + \underbrace{m_{ca} \cdot c_{ca}}_{\text{cage}} \quad (1)$$

$$C_{\text{tot}} = C_{UW} + C_{DW} \quad (2)$$

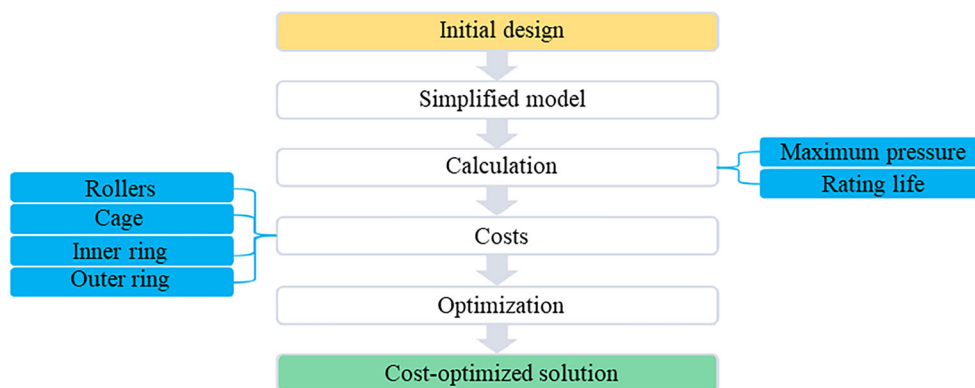
### 3.3 Optimization process

The optimization process is illustrated in Fig. 4, where the technical and economic evaluations are merged to identify the most cost-effective bearing configurations. The analysis started with an initial baseline design and gradually refined the geometric parameters, taking into account both performance metrics (such as maximum pressure and rating life) and cost factors. For each design, the following steps were performed:

- **Initial Design:** A simplified model of the main bearing unit was created to evaluate technical parameters such as maximum raceway pressure and rating life.



**Fig. 4** Principles of Design-to-Cost optimization process for bearing design



- **Cost Evaluation:** The cost of manufacturing the bearing was calculated by summing up the costs of each component (rollers, inner and outer rings, cages) using the mass-specific cost model. The cost optimization process was iterative, ensuring that each technical design adjustment was analyzed for its impact on both performance and cost.
- **Final Optimization:** The optimal bearing design was determined by evaluating multiple configurations, aiming to achieve the best compromise between minimizing contact pressure, extending rating life, and reducing manufacturing costs. The objective was to identify the most cost-effective variant that still met all technical requirements for pressure and rating life, ensuring both high performance and lower overall system costs.

## 4 Results

The results of this study provide a comprehensive analysis of the TRB-TRB main bearing system, focusing on the relationship between roller diameter, load angle, load distribution, contact pressure, and cost efficiency. Through a series of simulations and cost analyses, the impact of varying design parameters on bearing performance was systematically evaluated. This section presents the key findings, which include technical performance metrics and the corresponding cost implications. Figure 5 visually represents the examined combinations of roller diameter and load angle for the upwind bearing, with yellow dots indicating the analyzed configurations. Moreover, the extreme design scenarios—large/small roller diameter and high/low load angle—are schematically depicted to emphasize the differences between these design extremes. Throughout the study, the outer and inner diameters of both the upwind and downwind bearings were held constant, ensuring consistency and allowing a focused evaluation of the other design factors.

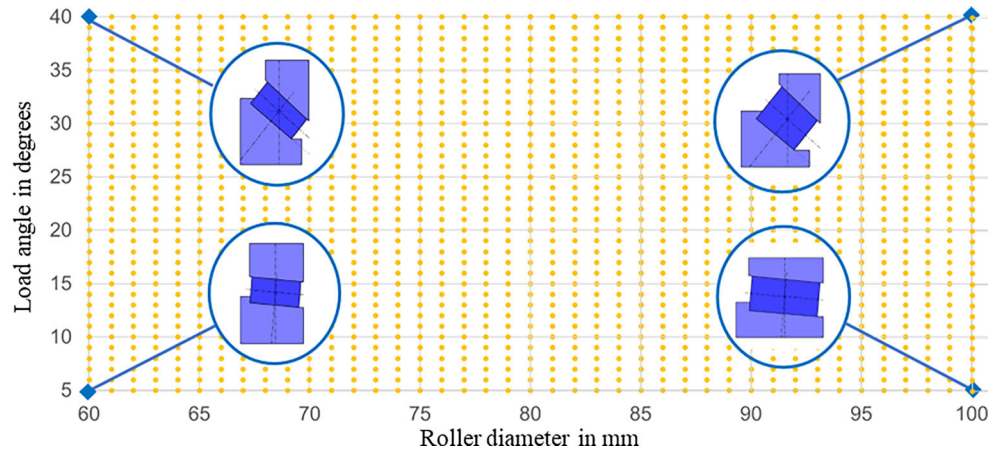
### 4.1 Load distribution and contact pressure

One of the primary objectives of this study was to evaluate how changes in roller diameter and load angle affect the load distribution and contact pressure within the bearings. By varying roller diameters between 60 and 100 mm and adjusting load angles from 5 to 40°, the simulation results reveal clear trends in how these parameters influence bearing performance. The analysis shows that increasing the roller diameter reduces the maximum contact pressure. As depicted in Fig. 7 (top), configurations with larger roller diameters (particularly between 77 and 94 mm) and moderate load angles (approximately 25 to 35°) resulted in the lowest contact pressure values (<1200 MPa). It should be noted that this diagram, along with the subsequent ones, illustrates variations in the upwind (UW) geometry while keeping the downwind (DW) geometry constant.

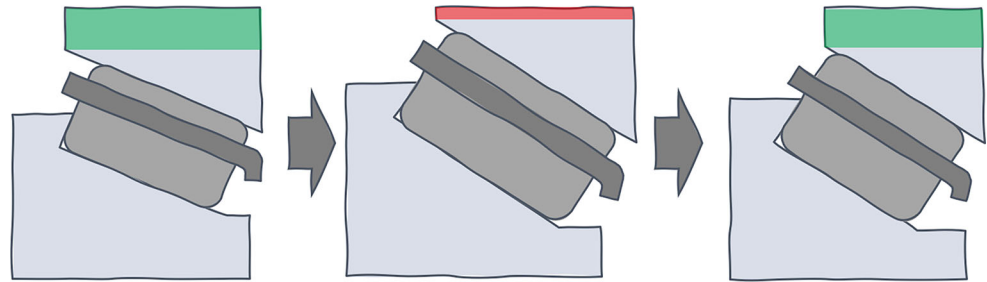
### 4.2 Bearing rating life

The impact of roller diameter and load angle on the bearing's rating life was also a key focus of the study. Using ISO standards for reference rating life (L10r) calculations, the results indicate that configurations with larger roller diameters and optimized load angles (particularly between 25 and 35°) significantly increase the expected operational life of the bearing. Figure 7 (top) illustrates the relationship between these parameters and bearing life. Configurations with roller diameters between 75 and 100 mm, paired with load angles of 25 to 35°, demonstrated rating life estimates well beyond the 20-year mark, which is a critical requirement for modern wind turbine applications. In contrast, configurations with smaller roller diameters and extreme load angles (above 35 or below 10°) showed significantly shorter life ratings due to higher contact pressures and less efficient load distribution. These results underscore the importance of optimizing both roller diameter and load angle to maximize bearing life and, by extension, the reliability of the wind turbine system.

**Fig. 5** Evaluated combinations and illustrative design configurations (upwind)



**Fig. 6** Geometric boundary restrictions for roller length with increasing roller diameter. The length is limited by restriction of the outer ring. (example)



### 4.3 Cost analysis and optimization

Alongside the technical performance evaluation, an in-depth cost analysis was conducted to determine the economic viability of different design configurations. By applying a mass-specific cost model, the study explored how changes in roller diameter, load angle, and component mass impact manufacturing costs. The findings suggest that increasing roller diameters generally leads to a notable rise in costs. This is primarily driven by limitations on the L/D (length-to-diameter) ratio of the rollers (the maximum is set to 1.5), which allow longer rollers and a wider bearing as the roller diameter increases. However, geometric and material constraints on the outer and inner diameters create a counterbalancing effect, resulting in adjustments to the bearing width, as illustrated in Fig. 6. Steep load angles further amplify this restriction, intensifying the cost effect. Finally, the analysis shows that two vastly different design approaches can yield comparable overall costs, as depicted in Fig. 7 (top).

### 4.4 Optimization of bearing geometry

The study's final stage focused on identifying the most cost-optimized bearing geometry based on the combined results of the technical and cost analyses. Figure 7 (middle) presents a clear illustration of the optimization process, showing how adjustments to roller diameter and load angle

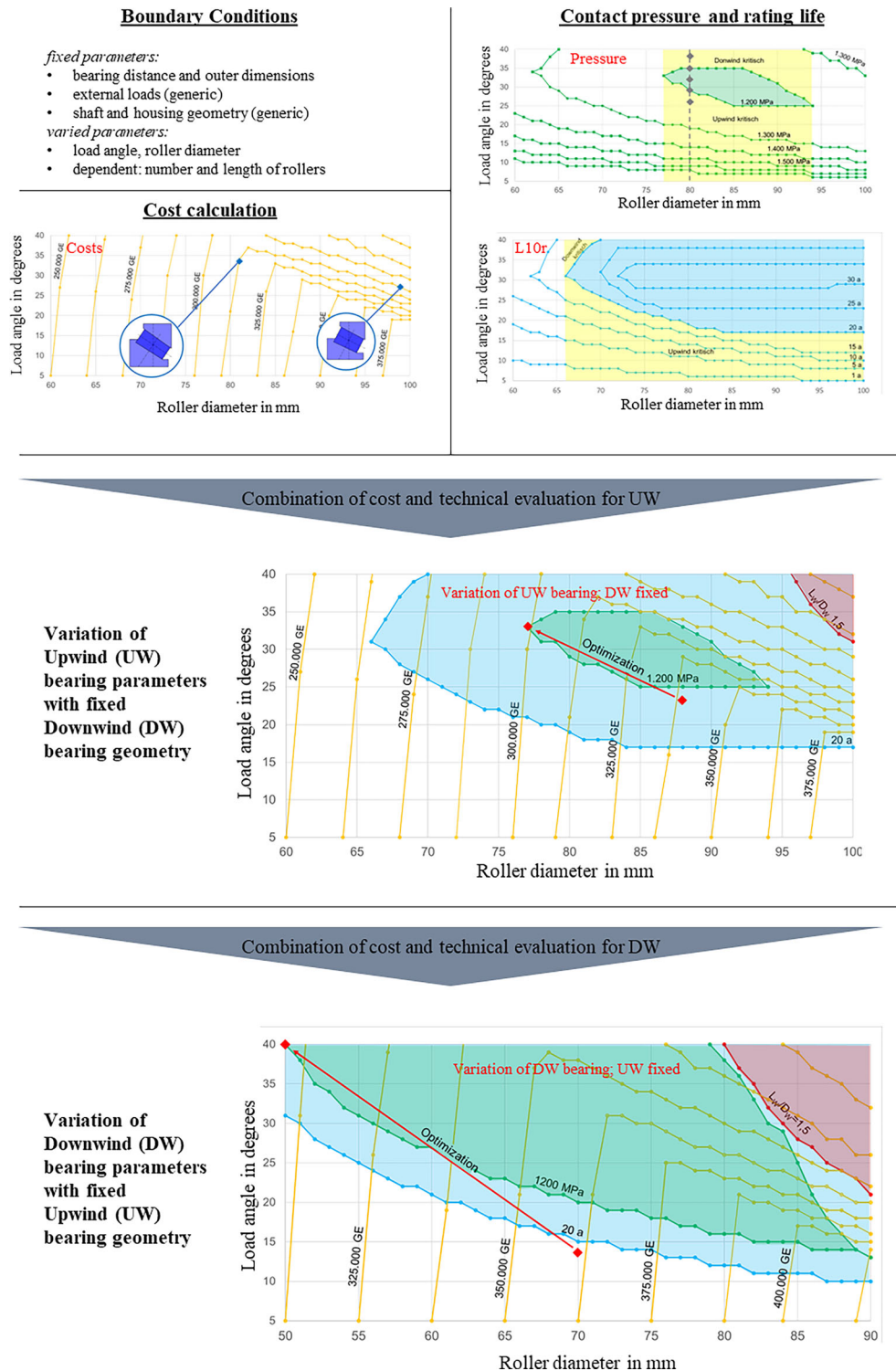
can lead to a bearing design that balances all performance criteria. Key findings include:

- **Technical Optimization:** Larger roller diameters (77 to 94 mm) combined with load angles between 25 and 35° offered the best performance in terms of reducing contact pressure and extending bearing life (green area).
- **Cost Optimization:** Within these configurations the most cost-efficient, with reduced material usage, leading to lower manufacturing costs and simplified production processes, is the one with smallest roller diameter.
- **Critical Thresholds:** Configurations with roller diameters above 94 mm or load angles exceeding 35° did not provide additional performance benefits and, in some cases, led to higher costs due to increased material demands and manufacturing complexity.

An optimization of the initial design resulted in a significant cost reduction, decreasing the total cost from approximately 400,000 to 337,500 monetary units (GE), which equates to a cost saving of around 15%. This reduction was achieved by refining the bearing geometry to optimize material usage while maintaining the required technical performance.

The same optimization procedure was applied to the downwind bearing, this time keeping the geometry of the upwind bearing fixed. As shown in Fig. 7 (bottom), the cost of the original downwind bearing design was reduced from approximately 360,000 to 310,000 GE, representing

**Fig. 7** Top: Results of technical and cost calculation. Middle and bottom: Optimization of UW (middle) and DW (bottom) bearing



a cost saving of around 14%. This optimization was similarly achieved by refining the bearing geometry while ensuring that critical technical parameters, rating life and contact pressure, were maintained.

It is important to note that this analysis primarily focused on optimizing rating life and pressure, while other relevant bearing characteristics—such as kinematics, shaft misalign-

ment, and other dynamic performance factors—were kept out of scope for this study. Future work may consider these aspects to provide a more comprehensive assessment of bearing performance.



## 5 Discussion and conclusion

The findings of this study offer valuable insights into the relationship between bearing design parameters and wind turbine performance, particularly in achieving cost-efficient and reliable operation. The TRB-TRB configuration analyzed in this paper underwent a series of simulations, enabling a thorough investigation of how variations in roller diameter, load angle, and bearing geometry affect critical technical factors, such as load distribution, contact pressure, and rating life.

A key takeaway from this study is the potential to reduce future production costs by strategically exploring various bearing solutions early in the design phase and considering their cost implications. Beyond technical performance, the research emphasized the economic impact of bearing design optimization. By applying a mass-specific cost model, the study demonstrated how changes in component mass and material usage influence manufacturing costs. Optimizing the bearing system's geometry, particularly by reducing the mass of key components such as rollers, cages, and rings, leads to significant cost savings. This is especially important for larger wind turbines, where material expenses account for a substantial share of total CaPex. The TRB-TRB system, through optimized component mass, presents a practical pathway to reducing upfront costs while ensuring operational reliability. The study's use of a "Design-to-Cost" approach highlights the importance of early-stage cost analysis in wind turbine development. By integrating cost considerations at the design phase, engineers can develop more efficient designs that meet performance criteria and stay within budget constraints. This approach allows for a more comprehensive evaluation of turbine components, ensuring technical performance, material efficiency, and manufacturing costs are all addressed together.

While the economic analysis demonstrated significant potential for reducing production costs through bearing design optimization, the trade-off between CaPex and OPex must be carefully weighed, especially in wind turbine applications. For offshore turbines, OPex considerations and safety against bearing failure are particularly critical due to the high costs of maintenance operations and the logistical challenges of performing repairs in harsh marine environments. Ensuring long-term reliability and minimizing operational risks should be prioritized in these applications, even at the expense of slightly higher initial costs. In contrast, for onshore turbines and cost-sensitive markets, lowering CaPex while avoiding unacceptable concessions in reliability offers a valuable strategy. Companies could improve their competitiveness in those markets by focusing on cost-optimized designs that maintain acceptable performance standards. By addressing cost pressures without compro-

mising reliability, these companies could secure a stronger position in highly competitive global markets.

Although the shaft and housing were kept unchanged during the optimization process, variations in the contact angle and roller size of course can have significant qualitative effects on stresses and deformations in these components. It should be noted that the detailed analysis of shaft and housing stresses and deformations was beyond the scope of this work. However, variations in load angles directly influence the supporting distance  $L_1$  (as shown in Fig. 2), which in turn impacts system-level parameters such as shaft misalignment and system stiffness. Similarly, changes in roller diameter can alter the effective load distribution and stiffness, indirectly affecting the shaft and housing integrity. Furthermore, changes in roller diameter and load angle influence the mechanical integrity of the bearing rings. The inner ring, which is clamped with a press fit and subjected to hoop stress, is particularly sensitive to these variations—if the ring is too thin due to increased roller size or modified load angles, it could compromise the structural stability of the assembly. Creeping, a phenomenon where relative motion occurs between the bearing rings and shaft or housing under load, is also affected by both bearing geometry and shaft/housing design. Larger rollers or increased load angles may alter the contact dynamics, potentially exacerbating creeping under certain conditions. These interactions highlight the necessity for a holistic approach to bearing and drivetrain design, considering the interdependence of components to maintain system reliability.

While the results are promising, there are areas for further research. While the study focused primarily on two technical parameters—bearing rating life and contact pressure—the approach could be expanded to incorporate additional relevant factors. Furthermore, this study focused mainly on onshore wind turbine operations. Offshore turbines, which face more challenging environmental conditions, may require additional considerations for future optimizations. In summary, this study provides a clear framework for optimizing wind turbine main bearing systems, showing that careful attention to design parameters such as roller diameter and load angle can result in significant improvements in both performance and cost efficiency. As the wind energy sector continues to expand, these findings will support the ongoing efforts to make renewable energy more affordable, accessible, and reliable on a global scale.

**Funding** This study was conducted and financed by thyssenkrupp rothe erde Germany GmbH, Lippstadt, Germany.

## References

1. Carroll J, McDonald A, Dinwoodie I, McMillan D, Revie M, Lazakis I (2017) Availability, operation and maintenance costs of off-

- shore wind turbines with different drive train configurations. *Wind Energy* 20(2):367–378
2. Ehrlenspiel K, Kiewert A, Lindemann U, Mörtl M (2020) *Kostengünstig Entwickeln und Konstruieren*, 4th edn. Springer, Berlin
  3. Frumușanu G, Cezarina A, Păunoiu V (2020) Estimation of Roller Bearings Manufacturing Costs by Causal Identification and Comparative Assessment—Case Study Performed on Industrial Data. *Int J Model Optim* 10(4)
  4. Guo Y, Parsons T, Dykes K, King RN (2016) A systems engineering analysis of three-point and four-point wind turbine drivetrain configurations. *Wind Energy* 20(3):537–550
  5. Harzendorf F, Azzam B, Schelenz R, Jacobs G (2018) Manufacturing cost—a critical evaluation criteria for new developments in wind turbine drivetrain technologies. *Series: J Phys* 1102:
  6. Harzendorf F, Schelenz R, Jacobs G (2020) Method for holistic wind turbine drivetrain comparison exemplarily applied to geared and direct drive systems. *Forsch Ingenieurwes* 86:21–33
  7. McMillan D, Ault GW (2010) Techno-Economic Comparison of Operational Aspects for Direct Drive and Gearbox-Driven Wind Turbines. *Ieee Trans Energy Conversions* 25(1)
  8. Polinder H, van der Pijl FFA, de Vilder G-J, Tavner PJ (2006) Comparison of Direct-Drive and Geared Generator Concepts for Wind Turbines. *Ieee Trans Energy Convers* 21(3)
  9. Reder MD, Gonzales E, Melero JJ (2016) Wind Turbine Failures—Tackling current Problems in Failure Data Analysis. *J Phys: Conf Ser* 753:72027
  10. Stehly T, Beiter P (2020) 2018 Cost of Wind Energy Review. National Renewable Energy Laboratory, Golden, CO, pp 5000–74598

**Publisher's Note** Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.